Maatra

HEALTH SUITE · VERNACULAR MEDICINE & DOSE COMPANION

BOSSCODER SCHOOL OF TECHNOLOGY HACKATHON 2026

Theme : Vernacular Medicine Visual Dose Companion

The Problem : Small English Labels make it easy to miss doses or confuse similar pills.Maatra makes routines visual and reminders speak the user's language.

Team - 404:Boys Not Found

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01 · PROBLEM STATEMENT AND NEED

Problem:People don’t remember the dosage

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Challenges:

- Missed Doses: Patients forget scheduled medications.
- Pill Confusion: Similar-looking medicines are difficult to distinguish.
- Dosage Errors: Small, English-only packaging makes instructions hard to understand.
- Language Barriers: Existing instructions may not be accessible in local languages.
- Caregiver Burden: Managing medications for seniors and multiple children is challenging.

Solution:

-Visual Medicine Setup: Identify pills using colour, shape, and dosage icons.
-Personalized Routine: Set medication schedules and dosage reminders.
-Vernacular Voice Alerts: Deliver spoken reminders in the user's preferred language.
-Dose Tracking: Track taken, upcoming, and missed doses.
-Caregiver Support: Simplify medication management for families and senior citizens.

Uniqueness:

-Visual-First Design: Uses pill colours, shapes, and dosage icons instead of relying only on text.
-Local-Language Voice:Converts reminders into spoken alerts in the user's preferred language.
Inclusive Experience: Designed for senior citizens, parents, and users with limited literacy.
-Personalized Reminders: Combines individual schedules with understandable visual and audio cues.

Benefits:

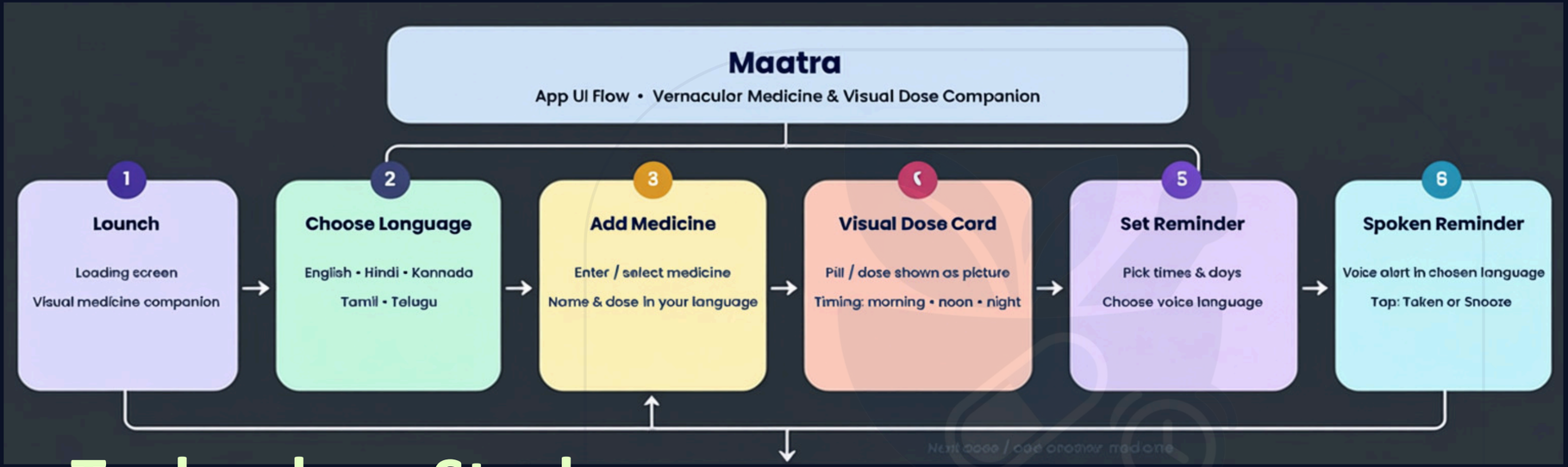
-Improved Adherence: Helps users follow prescribed medication schedules.
Reduced Confusion: Visual cues make medicines easier to identify.
-Better Accessibility: Supports users who struggle with English instructions.
-Greater Independence: Enables seniors to manage routines more confidently.

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Medicine Dosage and Reminder Companion

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02 · App Workflow

User Interface:



Technology Stack:



Methodology

- Collect and organize official BIS standards and service documents.
- Build a centralized, searchable knowledge base.
- Integrate AI, NLP, and RAG to understand queries and generate accurate answers.
- Provide multilingual chatbot support for easy access.
- Display answers with official references and links.
- Test the system for accuracy, reliability, and usability.

03 · Practicability

Feasibility ,Viability:

Feasibility

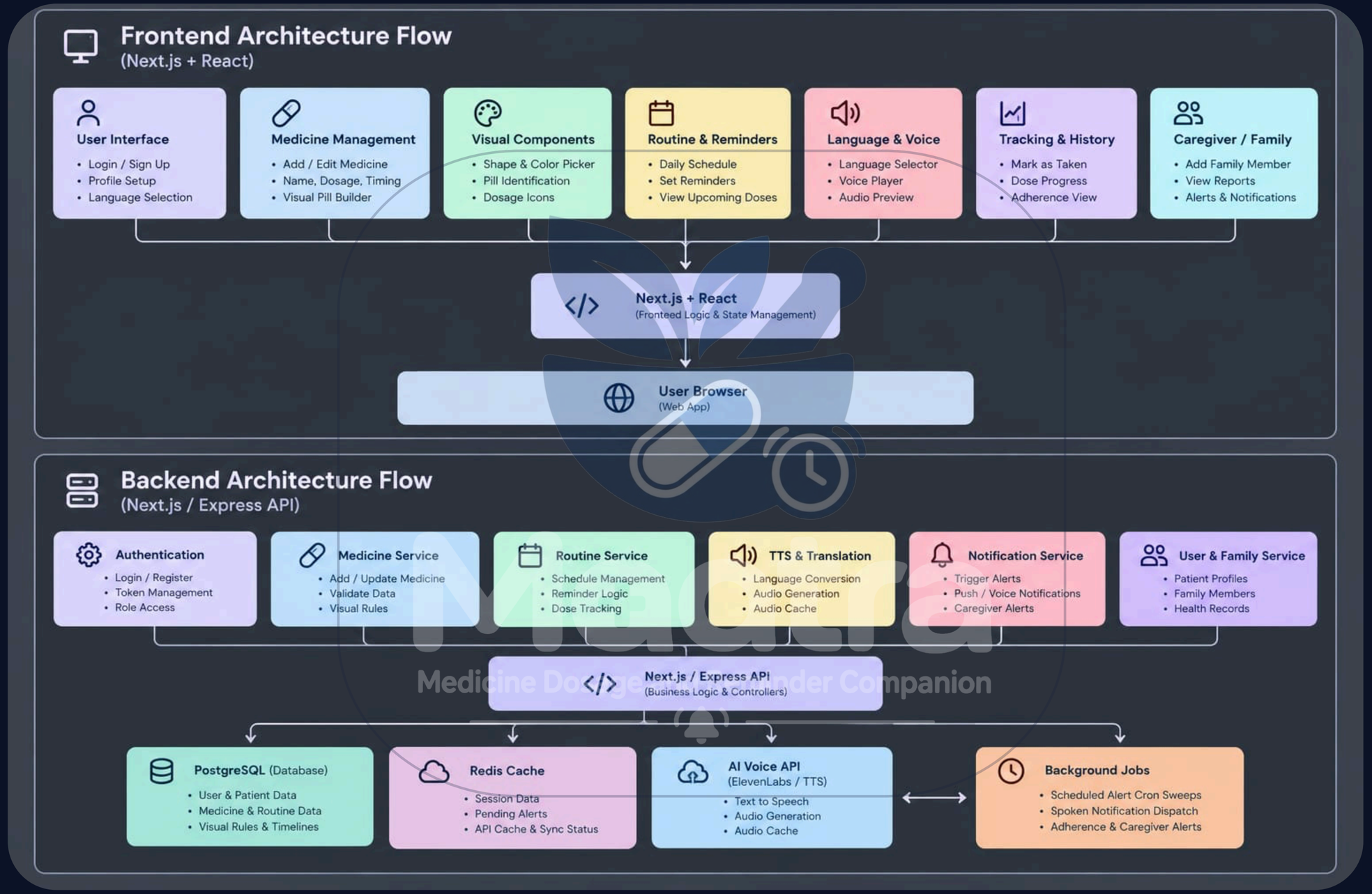
- Technical Feasibility: Uses AI/ML algorithms, databases, and web technologies for missing-person matching.
- Operational Feasibility: Simple interface for reporting missing persons and tracking case status.
- Data Feasibility: Stores missing-person details, photos, and last-seen locations securely.
- Implementation Feasibility: Can be developed using React, Node.js, and MongoDB/PostgreSQL.

Viability

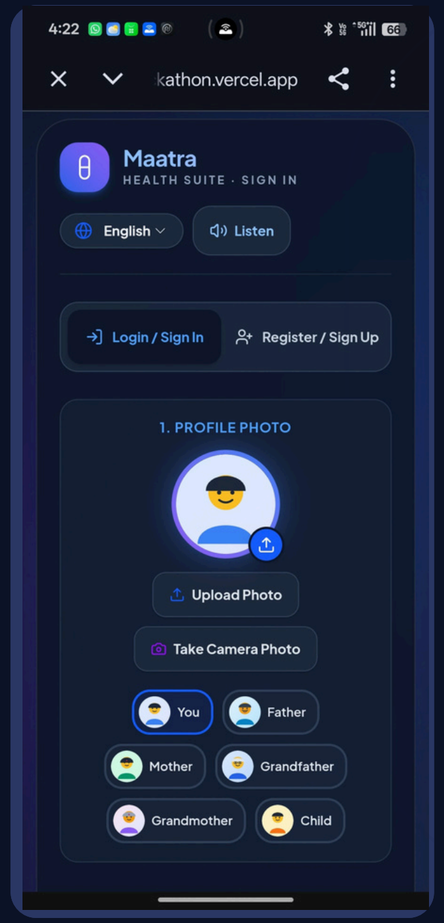
- Cost-Effective: Can be built using open-source technologies and cloud services.
- Scalability: Can expand to support more users and reports.
- Social Impact: Helps families and authorities find missing people.
- Practicality: Automates report matching and notifications to relevant people.
- Sustainability: Can collaborate with NGOs, police, and community organizations.

Strategies

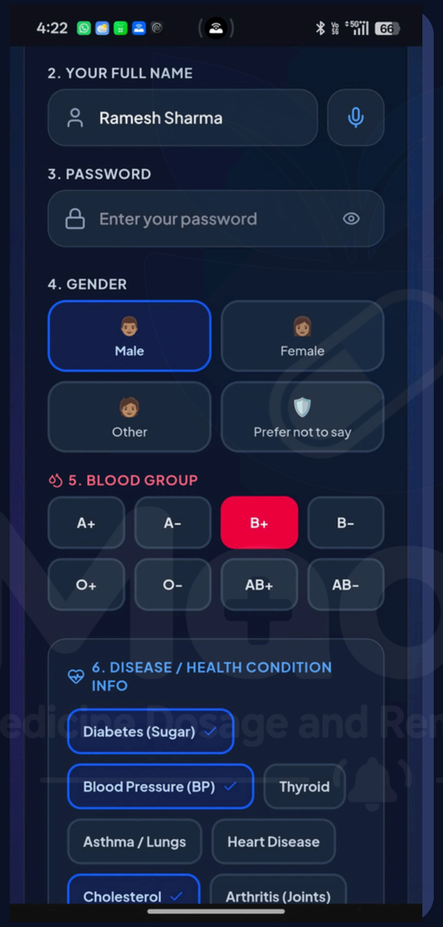
- Data Privacy: Encrypt personal information and restrict access to authorized users.
- Image Matching Difficulties: Use image-processing techniques and human verification.
- Notification Delays: Integrate reliable email, SMS, and alert systems.
- Technical Issues: Perform regular testing, backups, and system maintenance.
- User Accessibility: Provide a simple interface with multilingual support.


Frontend Architecture Flow
(Next.js + React)
User Interface
• Login / Sign Up
• Profile Setup
• Language Selection
Medicine Management
• Add / Edit Medicine
• Name, Dosage, Timing
• Visual Pill Builder
Visual Components
• Shape & Color Picker
• Pill Identification
• Dosage Icons
Routine & Reminders
• Daily Schedule
• Set Reminders
• View Upcoming Doses
Language & Voice
• Language Selector
• Voice Player
• Audio Preview
Tracking & History
• Mark as Taken
• Dose Progress
• Adherence View
Caregiver / Family
• Add Family Member
• View Reports
• Alerts & Notifications
Next.js + React
(Fronteed Logic & State Management)
User Browser
(Web App)
Backend Architecture Flow
(Next.js / Express API)
Authentication
• Login / Register
• Token Management
• Role Access
Medicine Service
• Add / Update Medicine
• Validate Data
• Visual Rules
Routine Service
• Schedule Management
• Reminder Logic
• Dose Tracking
TTS & Translation
• Language Conversion
• Audio Generation
• Audio Cache
Notification Service
• Trigger Alerts
• Push / Voice Notifications
• Caregiver Alerts
User & Family Service
• Patient Profiles
• Family Members
• Health Records
Next.js / Express API
(Business Logic & Controllers)
PostgreSQL (Database)
• User & Patient Data
• Medicine & Routine Data
• Visual Rules & Timelines
Redis Cache
• Session Data
• Pending Alerts
• API Cache & Sync Status
AI Voice API
(ElevenLabs / TTS)
• Text to Speech
• Audio Generation
• Audio Cache
Background Jobs
• Scheduled Alert Cron Sweeps
• Spoken Notification Dispatch
• Adherence & Caregiver Alerts

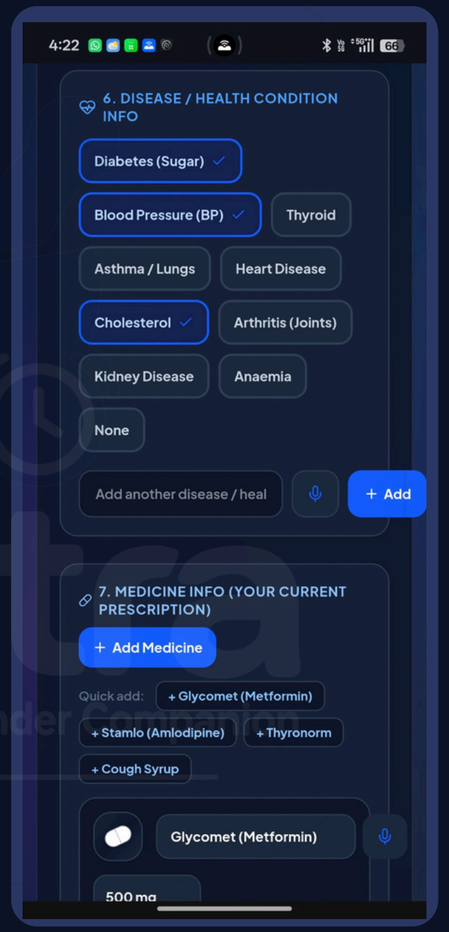
Prototype walkthrough: CLICK ME



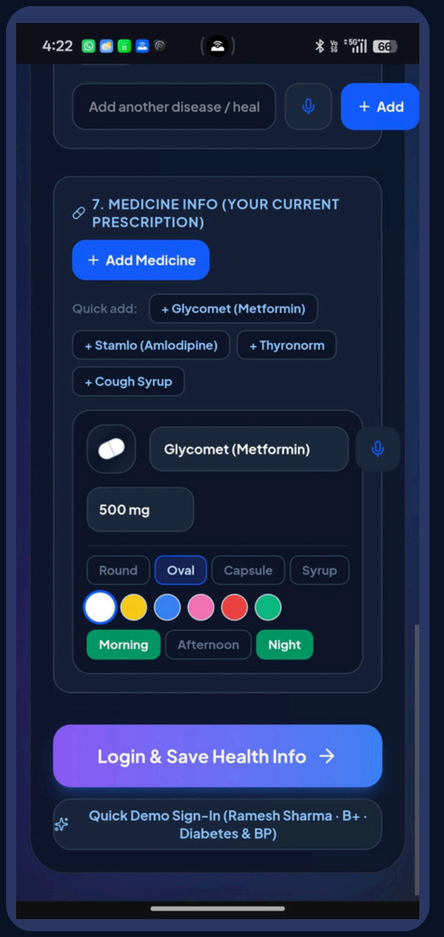
Language, Listen & family profile



Name by voice, gender, blood group



Health conditions & quick-add



Pill shape, colour & dose times

References:

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1 **WHO**- click for info

The WHO operates under a broad mandate to achieve the highest possible level of health for all people through several core responsibilities:

2 **Translator**- click for info

AI Language Model APIs (Most Common): The application likely makes API calls to models like OpenAI's GPT-4o or Google's Gemini via serverless functions (Next.js API routes).

3 **Bhashini** -click for info

Provides APIs for speech recognition, translation, text-to-speech, and OCR. These can help users speak medicine instructions, translate them, and hear reminders in their preferred language.

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